NeuroStream User Guide

Overview – What is NeuroStream?

NeuroStream is an interactive visualization tool designed to support exploratory data analysis (EDA) of regional brain features derived from neuroimaging pipelines (e.g., MRI-based processing).

The tool enables users to explore, without writing code**:** Distributional differences across cohorts, Changes in data characteristics under different preprocessing methods, Relationships between brain measures and clinical variables (e.g., age, years of education).

1. Data Upload

This screen allows users to upload the input dataset for analysis.

Supported format: CSV file

Data structure: One row per subject

cohort	age_group	gender	edu_level	htn	dm	icv_300	whole_bra	ctx_cerebr	cerebral_
Cohort1	3	2	0	1	2	1528.97	1087.181	475.854	456.194
Cohort1	3	1	2	1	1	1672.834	1155.588	510.907	468.693
Cohort1	1	2	1	2	2	1270.509	916.385	417.169	386.732
Cohort1	3	2	1	2	2	1606.304	969.974	438.748	412.385
Cohort1	3	2	1	1	2	1489.17	957.11	429.918	418.315
Cohort1	4	2	1	2	2	1247.569	857.186	387.895	359.236
Cohort1	3	1	6	1	2	1529.645	1083.463	466.072	464.204
Cohort1	3	2	1	2	2	1518.339	956.883	411.802	448.83
Cohort1	3	1	1	1	2	1480.643	1072.522	428.421	436.001
Cohort1	2	2	3	2	2	1422.969	999.328	456.327	425.394
Cohort1	3	2	3	1	1	1447.983	1042.655	487.187	476.371

Required contents:

· Cohort information

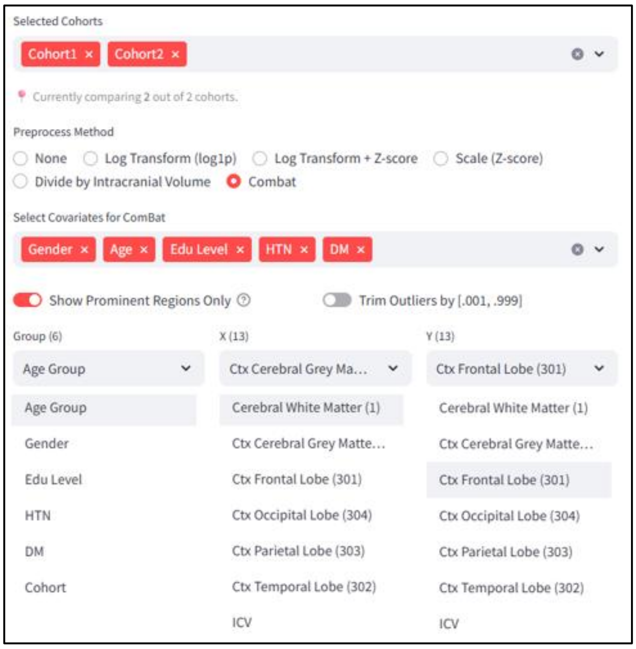
· Clinical variables (e.g., age, education level)

· Regional brain volumes

The uploaded dataset serves as the foundation for all subsequent visualizations and statistical summaries within NeuroStream.

2. Cohort and Preprocessing Selection

At this stage, users select the cohorts and preprocessing methods to include in the analysis.



· Cohort Selection: Multiple cohorts can be selected for group-wise comparison.

· Dynamic Preprocessing: Different methods can be applied to adjust data scale and distributional properties.

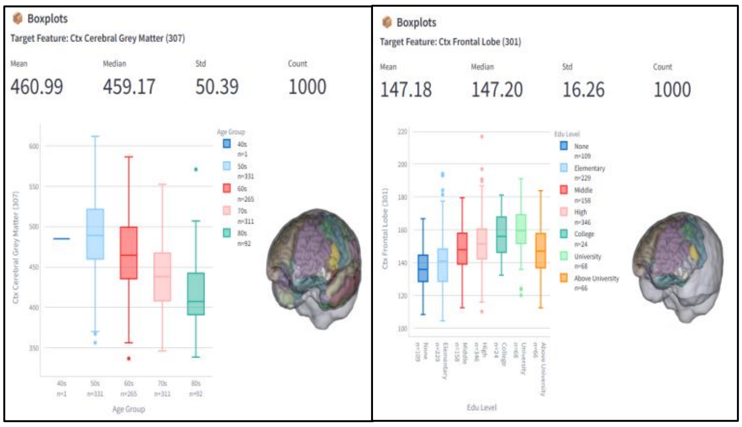
· Flexible ComBat Covariates: When 'Combat' is selected, users can now manually choose multiple covariates (e.g., Gender, Age, Edu Level, HTN, DM) to adjust for specific biological factors.

· Real-time Updates: Changes in cohorts or preprocessing (including covariate selection) are immediately reflected in all downstream visualizations, such as PCA, boxplots, and residual analysis.

3. Exploratory Data Analysis (EDA)

3.1. Boxplot, descriptive statistics

This view presents descriptive statistics and boxplot visualizations for the selected variables.



Reported statistics:

· Mean, Median, Standard deviation, Sample size

Boxplots summarize distributional differences across *categorical variables*

(e.g., age groups or education groups) and are useful for comparing central tendency and variability between cohorts.

3.2. Groupwise T-test

This section provides results from pairwise group-wise t-tests between selected groups.

Groupwise T-test: Ctx Cerebral Grey Matter (307)^X by Age Group

*** p<0.05, ** <0.01, *** <0.001, or Not Significant**

Note: Pair-wise t-tests are performed for multiple cohorts.

	50s	60s	70s	80s
50s	-	***	***	***
60s	17.826	-	***	***
70s	26.959	10.305	-	***
80s	45.205	29.894	18.813	-

Groupwise T-test: Ctx Frontal Lobe (301)^Y by Edu Level

*** p<0.05, ** <0.01, *** <0.001, or Not Significant**

Note: Pair-wise t-tests are performed for multiple cohorts.

	None	Elementary	Middle	High	College	University	Above University
None	-	***	***	***	***	***	***
Element	-5.568	-	***	***	***	***	***
Middle	-10.175	-4.17	-	***	***	***	***
High	-13.93	-7.878	-4.007	-	***	***	***
College	-18.52	-12.08	-8.386	-4.134	-	***	***
Universi	-22.835	-16.98	-13.956	-10.181	-6.892	-	***
Above U	-27.95	-22.459	-19.934	-16.449	-13.716	-6.769	-

· Pairwise comparison of mean differences between age or education categories

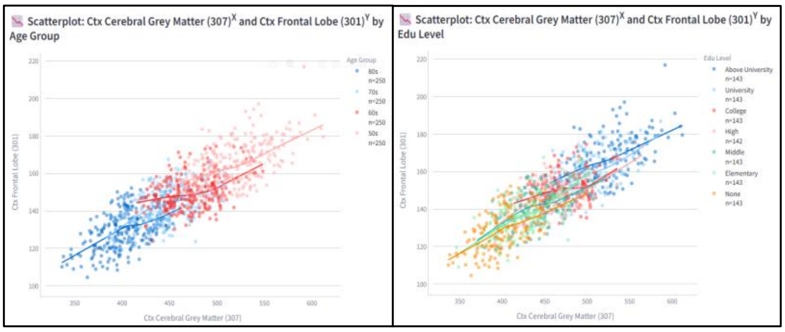
· Comparisons can be performed across cohorts or preprocessing conditions

This feature is intended for rapid exploratory assessment of potential group differences during early-stage analysis.

Note: NeuroStream is intended for exploratory data analysis and visualization. Statistical results are provided for descriptive purposes only and should not be interpreted as confirmatory inference.

3.3. Scatter plot

The scatter plot visualizes relationships between two continuous variables.



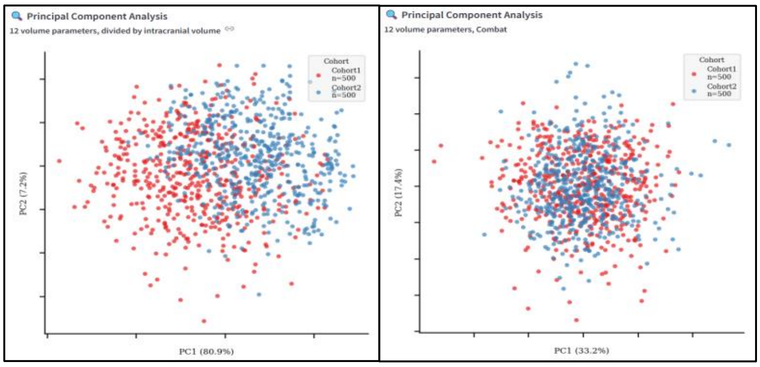
· Example: age vs. regional brain volume

· Points can be color-coded by cohort or group

This view enables intuitive inspection of trends such as age-related changes and cohort-specific patterns.

4. PCA (Principal Component Analysis)

PCA projects high-dimensional regional brain features into a lower-dimensional space for visualization.



· Overview of the global data structure

· Assessment of cohort-level separation or overlap

· Comparison of data structure before and after ComBat harmonization to visually assess reduction of batch effects.

4.1. Batch effect validation

This section provides comprehensive metrics and visualizations to verify the successful removal of site-specific artifacts (batch effects) after ComBat harmonization.

Statistical Metrics (Site Effect Evaluation)

: Provides quantitative evidence of harmonization performance across all brain features.

· Avg. Site p-value: Average ANOVA p-value across features. Values > 0.05 indicate no significant remaining site differences.

· Significant Features: Number of regions with $p < 0.05$. Ideal harmonization results in '0 / Total'.

· Mean F-statistic: Average variance magnitude between sites. Values closer to 0 reflect minimal inter-site bias.

Visual Check (Residual Analysis)

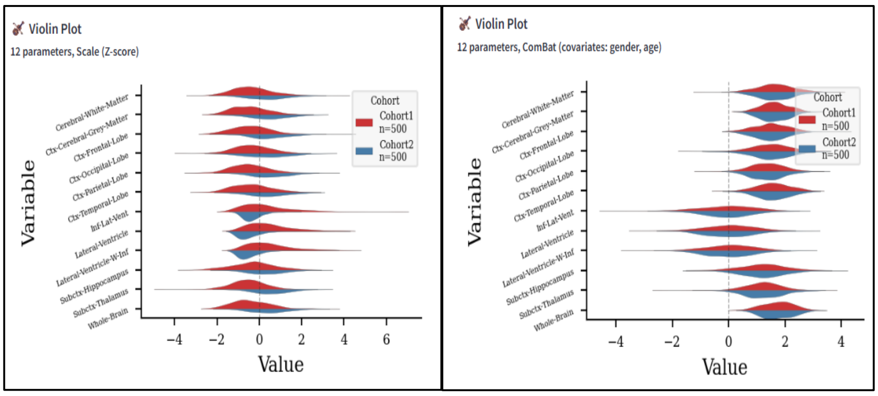
: Assesses the alignment of residual distributions for a selected feature to verify batch effect removal.

· Pre-Combat Residuals: Misaligned box plots across sites indicate existing batch effects despite covariate adjustment.

· Post-Combat Residuals: Box plots aligned along the zero line (red dashed) confirm that site biases have been removed.

4.2. Violin plot

Violin plots visualize the full distribution of regional brain measures using kernel density estimation.



· Provide more detailed distributional information than boxplots

· Useful for inspecting asymmetry and distributional spread

By comparing violin plots before and after ComBat harmonization, users can intuitively assess how harmonization affects the overall distribution of brain features across cohorts.